MATH317 · NUMERICAL METHODS

Midterm 472 Practice

12/04/2026 · 60 minutes · 30 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 TRUE / FALSE · 7 M

1) Significant digits are only important when rounding final results, not during intermediate calculations.
2) Horner's algorithm is used for efficiently evaluating polynomials.
3) The Taylor series converges rapidly for points far from the expansion point.
4) The Bisection method guarantees finding the exact root in a finite number of steps.
5) Rounding replaces a number with an approximation that minimizes error, whereas chopping discards digits without adjustment.
6) The Mean-Value Theorem is a special case of Taylor's Theorem with zero derivatives.
7) The error in the Bisection method after n steps is at most (b – a) divided by 2^n.

Q2 3 + 5 M

Given $f(x) = 22 - 4x^2 + 3x^3$.

a) Use Horner's algorithm to evaluate f(x) at x = −2.

b) What is the Taylor series for the function f(x) about c = −3?

Q3 7 M

Determine the single-precision machine representation in a 32-bit word-length computer of the decimal number (−23.21875).

Q4 2 + 6 M

Given $f(x) = -x^2 + 4x - e^x$.

a) Prove that f(x) has exactly one real root α in the interval [0, 1].

b) Find a root of f(x) on [0, 1] by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.